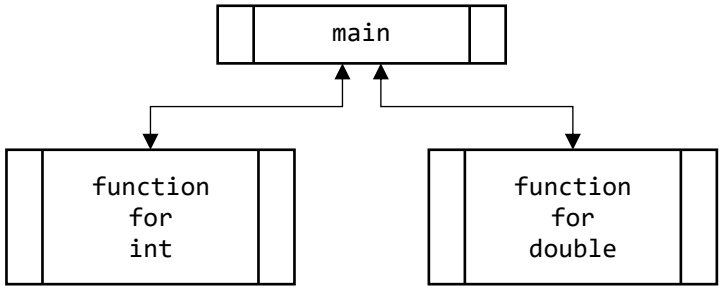
**ECE 2305 – Introduction to C Programming**
**Programming Tutorial 19**
**Overloaded Functions**

Write a C++ program to demonstrate user-defined functions and function calls as illustrated below. Both user-defined functions must have the same function name. This is an exercise in overloading the function name.




The functions are described below in "pseudo code."

function (return type void, argument type int, variable name *n*)
display " welcome to function for int"
display Memory Map for function
variable *n* address and value

function (return type void, argument type double, variable name *n*)
display " welcome to function for double"
display Memory Map for function
variable *n* address and value

main function:
function call with integer value
display "welcome back to main"
function call with double value
display "welcome back to main"

Document the exercise in a Word document.

**ECE 2305 – Introduction to C Programming**
**Programming Tutorial 19**
**Overloaded Functions**
**Input**

```
// ECE 2305 - Tutorial 19
// Overloaded Functions
#include<iostream>
using namespace std;

void function(int n)
{
	cout << "Welcome to function for int" << endl;
	cout << "n \t" << &n << "\t" << n << endl;
	cout << "Returning to main" << endl;
	cout << endl;
}

void function(double n)
{
	cout << "Welcome to function for double" << endl;
	cout << "n \t" << &n << "/t" << n << endl;
	cout << "Returning to main" << endl;
	cout << endl;
}

int main()
{
	cout << "ECE 2305 - Tutorial 19" << endl;
	cout << "Overloaded Functions" << endl;
	cout << endl;
	cout << "Executing Function main" << endl;
	cout << endl;
	cout << "Call function for int" << endl;
	cout << endl;
	function(100);
	cout << "Welcome back to main" << endl;
	cout << endl;
	cout << "Call function for double" << endl;
	cout << endl;
	function(3.14159);
	cout << endl;
	cout << "End of function main" << endl;
	cout << endl;

	system("pause");
	return 0;
}
```

**ECE 2305 – Introduction to C Programming**
**Programming Tutorial 19**
**Overloaded Functions**
**Output**

